

Cross-Track Contribution

Collaborating Track: Data Science

Collaborators: Donald Nwachukwu and Fatimah Odumuyiwa

Data Analytics provided:

- Week 5 No-Show findings
- Booking lead-time analysis
- Previous No-Show analysis
- Combined Lead Time $\times$ Previous No-Show segmentation
- Statistical validation results
- KPI refinement
- Key business insights requiring predictive validation

Data Science provided:

- Logistic Regression and Random Forest model results
- Model performance metrics
- Feature rankings and Logistic Regression coefficients
- Feature significance results
- Engineered feature validation
- Ablation study results
- Model error-analysis findings

How the collaboration changed the work:

The Data Science results strengthened the prioritisation of booking lead time and previous No-Show history while also identifying modelling redundancies among some engineered features. The ablation analysis showed that `booking_lead_days` provides more information than the grouped `lead_time_bucket`, while `is_new_patient` adds little beyond `previous_appointments`. It also showed that `distance_to_clinic_km` did not provide additional predictive value in the reported model setup despite its observed bivariate relationship in the Data Analytics analysis.

The collaboration therefore helped distinguish between variables that show descriptive associations and variables that add predictive value when considered alongside other features.

Cross-Track Integration Evidence

1. Track collaborated with:

Data Science Track

Collaborators: Donald Nwachukwu and Fatimah Odumuyiwa

2. Project dependency:

The Data Analytics findings required predictive validation to determine whether the factors identified through descriptive and statistical analysis remained useful when considered alongside other variables in predictive models.

3. Information/output received:

I received Logistic Regression and Random Forest results, model performance metrics, feature rankings and coefficients, feature significance results, engineered feature validation, ablation study results, and model error-analysis findings.

4. Information/output provided:

I provided the Data Science collaborators with my Week 5/6 findings on booking lead time, previous No-Show history, the combined Lead Time $\times$ Previous No-Show segment, distance, reminder variables, statistical validation results, KPI refinement, and key business insights.

5. Integration activity completed:

I compared the Data Analytics findings with the Data Science modelling results and reviewed the feature validation and ablation results to determine which analytical findings were also useful from a predictive-modelling perspective.

6. What changed as a result:

The collaboration strengthened the prioritisation of booking lead time and previous No-Show history, supported the combined Lead Time $\times$ Previous No-Show segment as a candidate feature for further modelling, and identified redundancies among some engineered features. It also showed that distance had limited additional predictive value in the reported model setup despite its observed bivariate relationship.

7. Evidence:

Evidence of the integration includes the Data Science model results, feature analysis, ablation study, error-analysis findings, exchanged analytical findings, and the documented changes incorporated into this Week 6 notebook, including KPI refinement, feature prioritisation, limitations, and recommendations.